Data Visualization Pitch

Consumer Behavior in Istanbul's Shopping Malls

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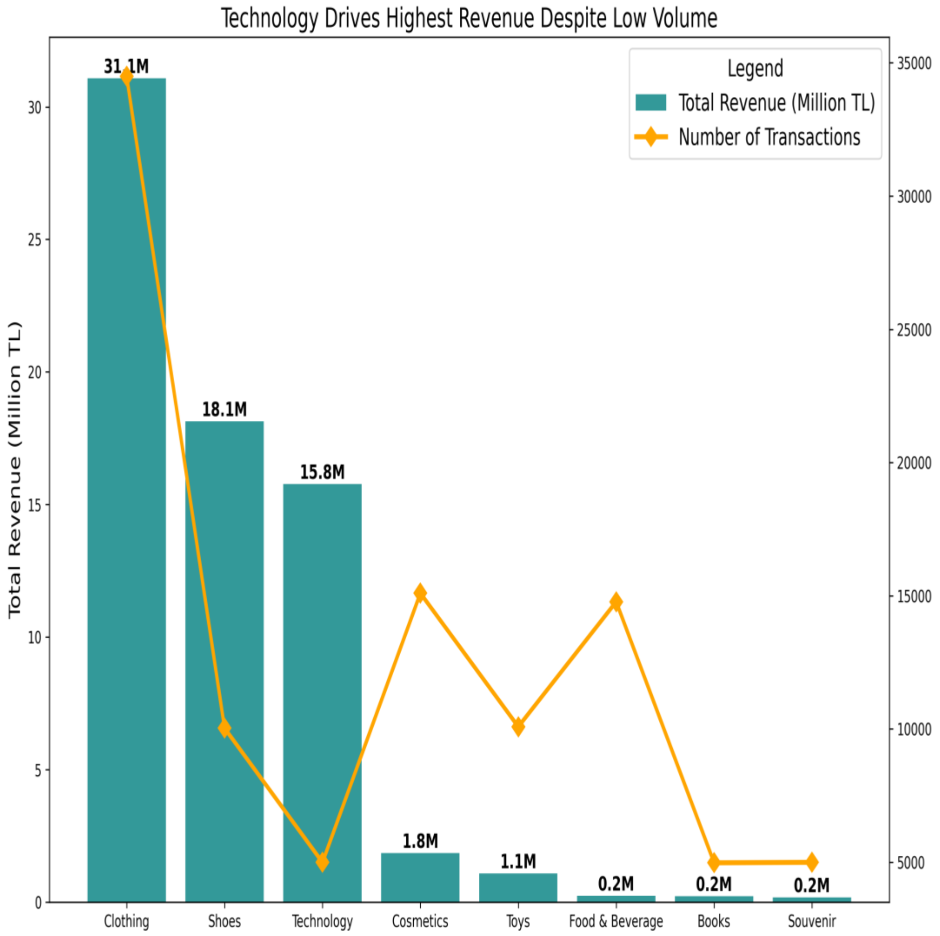
Student Individual Assignment



Unveiling Istanbul Shopping Trends: High-Value Categories and Mall Dominance (2021-2023)

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Research questions

What are the key drivers and patterns of consumer spending in Istanbul's shopping malls (2021–2023)?

- Which product categories generate the highest revenue, and how do they compare to transaction volume?
- Which shopping malls perform best in terms of sales volume and revenue?
- Is there a seasonal or holiday trend in shopping activity and spending?
- How do purchasing patterns differ across age groups and product categories in Istanbul’s shopping malls?

About Data

Data link → here

The **dataset** chosen for this analysis is “customer_shopping_data.csv,” which provides detailed records of 99,457 transactions from 10 shopping malls in Istanbul between 2021 and 2023. The data includes several attributes such as invoice numbers, customer IDs, gender, age, item categories (8 types), quantity, price (in TL), payment methods, invoice dates, and shopping malls. The data offers insights into consumer shopping behaviors, including real-world patterns such as holiday peaks.

While **searching for a dataset** on shopping trends in Istanbul, I found this dataset on Kaggle by searching "retail shopping trends Istanbul." The data collection itself was straightforward, and no major issues were encountered. I did consider alternative datasets, such as Eurostat tourism data, which could provide broader insights. However, this dataset stood out due to its focus on retail behavior within Istanbul's malls and its granular data on consumer behavior.

The dataset is **licensed** under the Creative Commons BY-SA 4.0 license, which allows for free use, redistribution, and adaptation, as long as proper attribution is given. There are no specific restrictions noted for this dataset, and it is open for analysis.

Concerning the quality and completeness of the data, while the dataset is largely complete and includes accurate transaction details, its scope is confined to 10 shopping malls in Istanbul. This regional limitation may restrict the findings’ applicability to broader retail markets in Turkey. Additionally, the dataset does not distinguish between tourist and resident shoppers, which is relevant in a city like Istanbul, where tourism can significantly affect seasonal spending patterns. Furthermore, all values are recorded in Turkish Lira (TL), and no currency conversion is applied, which might pose challenges for comparisons over time or across regions. Despite these limitations, the data's quality remains high for the intended analysis.

Regarding **data cleaning**, minimal operations were needed. I transformed the invoice_date column into a datetime format using Python to facilitate time-based analysis, checked for duplicate records (none were found), and added a new column to calculate the total price for each transaction (quantity × price) for revenue analysis.

Methodology

Data Collection: Downloaded CSV from Kaggle. No scraping/APIs needed.

Data Processing & Cleaning: I used Python with pandas to load and process the customer shopping dataset from Kaggle. First, I handled missing values, if any, and ensured correct date parsing by applying pd.to_datetime(). I verified that there were no duplicates in the dataset by cross-checking the invoice_no and customer_id columns. For data consistency, I also added a new calculated column total_price = quantity * price to help with revenue calculations. All records were checked for quality, and a few anomalies in the invoice_date were resolved to ensure the dataset was ready for analysis. Once the data was clean, I used Datawrapper to generate visualizations, creating clear insights into the revenue distribution across malls.

Data Transformation: After cleaning, I aggregated the revenue data by shopping mall using the groupby() function to analyze the total revenue for each mall. This transformation was crucial for comparing mall performance and identifying key revenue drivers. I also created time-based bins by extracting the month_year using dt.strftime('%Y-%m') to observe seasonal trends in consumer spending. To make the data more understandable, I normalized the revenue data by converting it to millions of TL and used Datawrapper to visualize these aggregated results, making it easier to communicate insights. I also grouped transactions by age groups and product categories to analyze demographic preferences and visualized this relationship using a heatmap.

Use of AI Tools: Grok (xAI) for code suggestions and initial aggregations (e.g., groupby queries); verified manually in Jupyter.

Analytical Techniques: Descriptive stats (sum/mean via groupby), trend analysis (monthly sorting), comparative analysis (revenue vs count), binning (age groups). Outlier detection: No major (prices within expected range).

Reproducibility: Fully replicable. Code shared in Jupyter notebook (attached on GitHub link); steps: Download CSV from Kaggle, run pandas groupby for aggregates, export to CSV for viz tools.

Insights from the Data

- **Mall Concentration of Revenue:** Mall of Istanbul and Kanyon together generate over 100 million TL, accounting for around 40% of total revenue among the 10 malls. The next group (Metrocity, Metropol AVM, Istinye Park) shows a noticeable drop in revenue compared to the top two. This indicates a strong dominance of flagship malls, suggesting that consumer spending is not evenly distributed across the city.

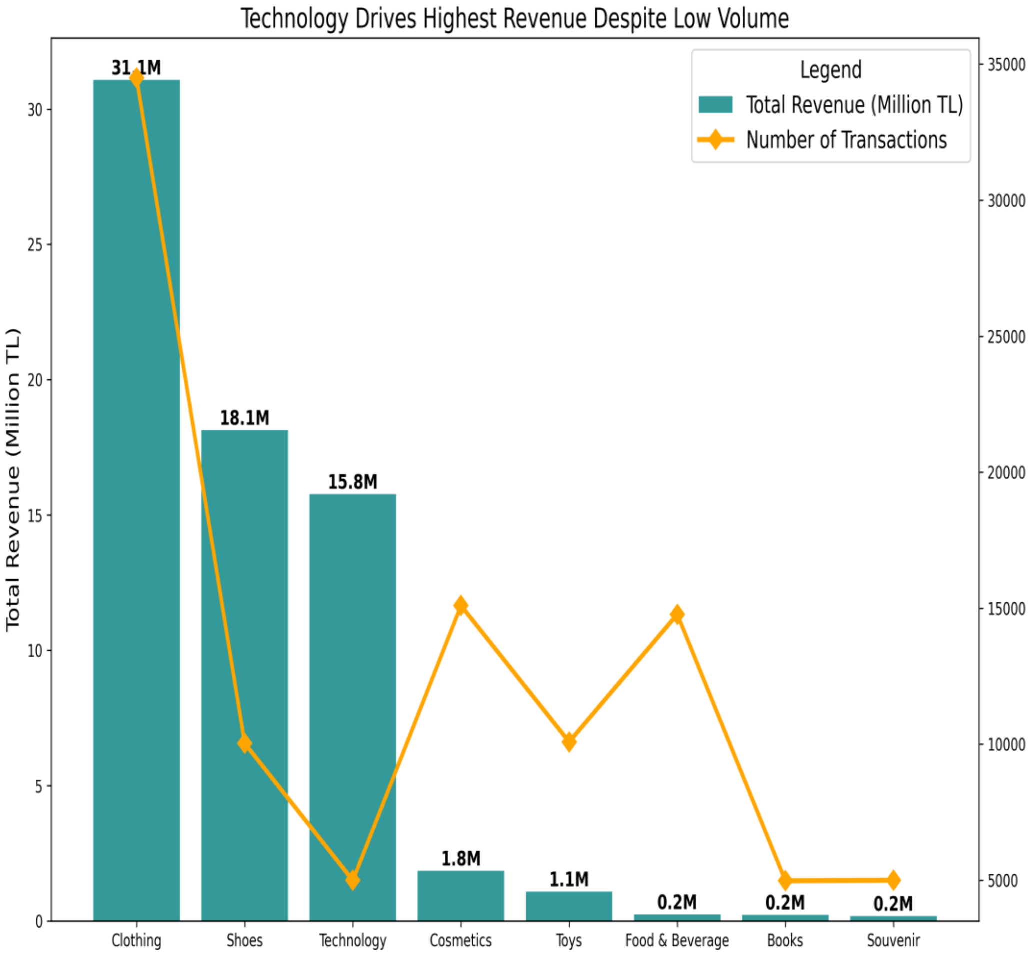
- **Category Revenue vs Transaction Volume:** Clothing shows the highest number of transactions, indicating it is the most frequently purchased category. However, Technology generates the highest total revenue despite having far fewer transactions. This suggests that average spending per transaction is much higher in technology than in clothing or shoes.

- **Seasonal Spending Patterns:** Monthly revenue trends show repeated peaks during specific periods of the year. Higher revenue is observed around holiday seasons and major shopping periods, such as year-end and back-to-school months. Lower revenue months indicate reduced discretionary spending outside promotional seasons.

- **Age-Based Purchase Preferences:** The age–category heatmap shows that clothing is the most frequently purchased category across all age groups, but purchasing intensity increases significantly in the 56+ group. Older consumers also show higher engagement in cosmetics, food & beverage, and toys compared to younger groups, suggesting more diversified shopping baskets among senior shoppers, while younger age groups concentrate more heavily on clothing and footwear.

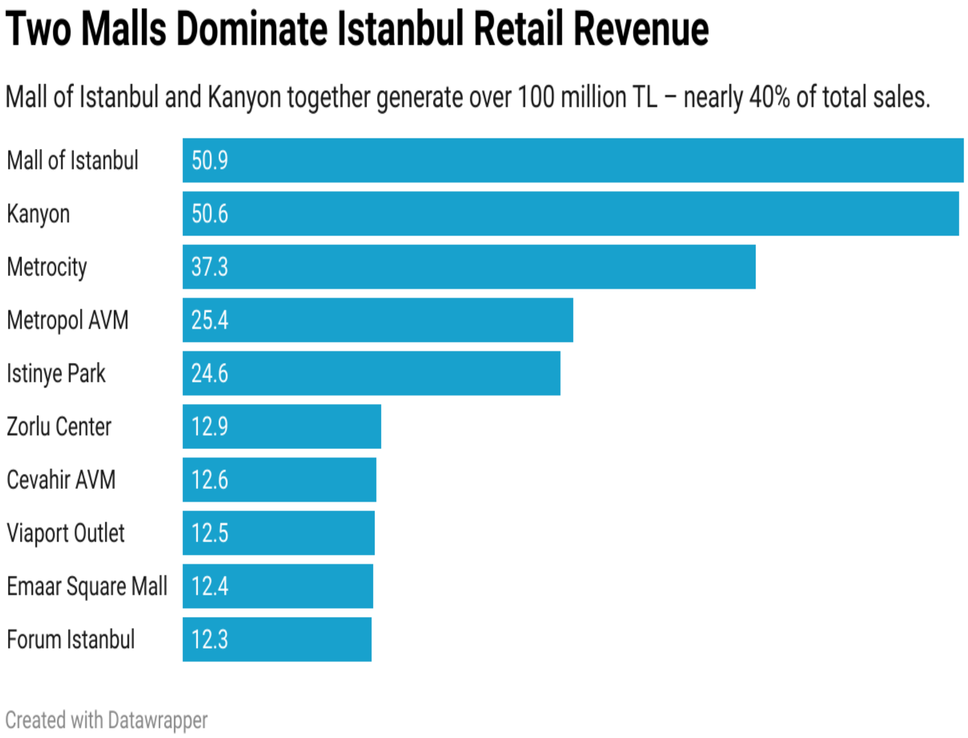
- **Analysis operations:** *Aggregation:* Grouped transactions by shopping mall, product category, and age group to compute total revenue and purchase frequencies. *Calculation:* Created a total_price variable (quantity × price) to measure actual spending per transaction. *Comparison:* Compared revenue and transaction volume across categories and malls to detect value–volume mismatches. *Time-series aggregation:* Aggregated data by month to analyze seasonal spending patterns.

Technology Drives Highest Revenue Despite Low Volume



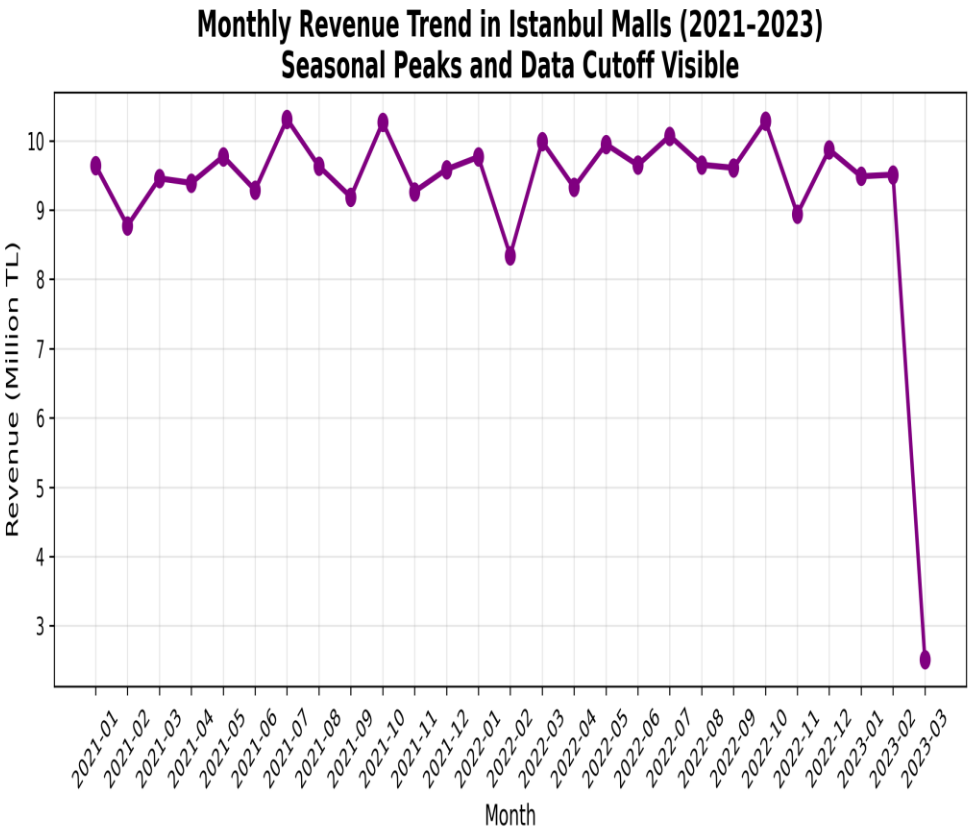
- The gap between revenue and transaction volume reveals a clear decoupling between how often products are bought and how much money they generate, with technology acting as a low-frequency but high-impact revenue driver.
- This value–volume mismatch suggests that different categories require different retail strategies: high-volume categories such as clothing benefit from promotions and fast turnover, while high-value categories like technology benefit more from premium placement, product variety, and customer service that supports higher ticket purchases.
- From a business perspective, this suggests that mall managers and retailers should not rely solely on foot traffic indicators, but also prioritize category mix and product value when designing tenant strategies and promotional campaigns.

Two Malls Dominate Istanbul Retail Revenue



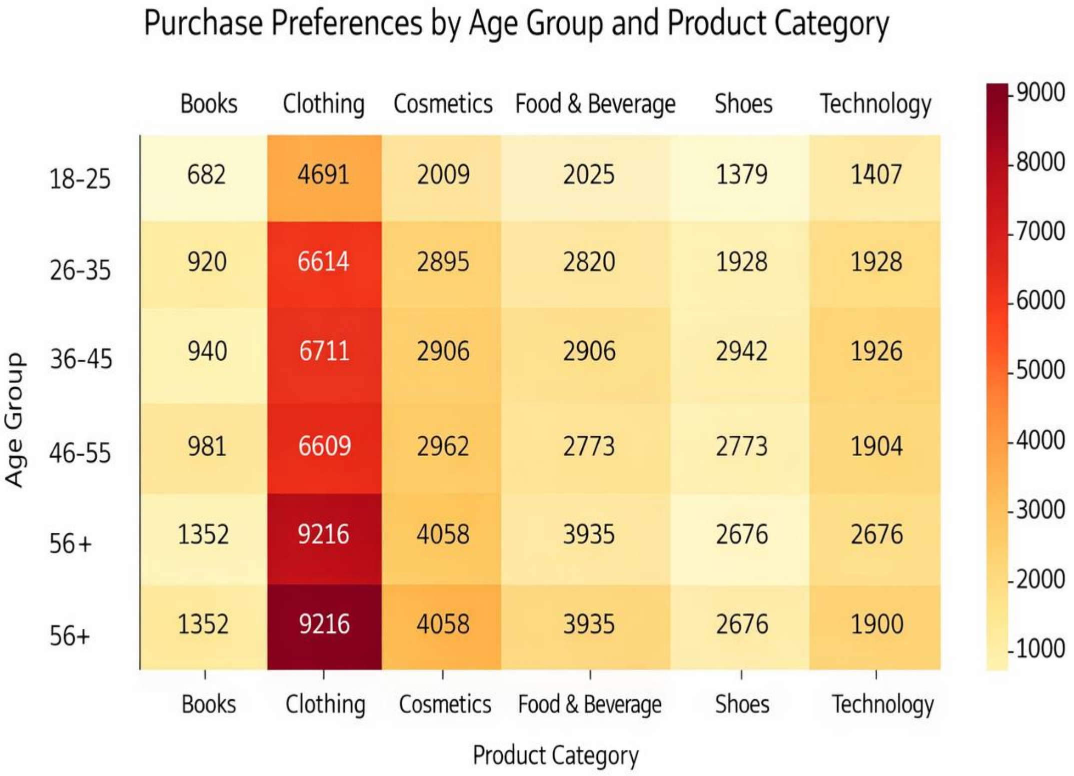
- The bar chart shows a highly uneven distribution of revenue across malls. Mall of Istanbul (50.9M TL) and Kanyon (50.6M TL) together generate more than 100 million TL in revenue, accounting for nearly 40% of total sales, while the remaining eight malls individually contribute less than half of that amount.
- This strong concentration suggests a “winner-takes-most” retail structure, where flagship malls benefit from cumulative advantages such as higher foot traffic, premium brand presence, and better transport accessibility. For smaller malls, this implies that competing on scale alone may be ineffective, and targeted promotions or niche positioning may be more viable strategies.

Revenue trends show repeated peaks during the holiday seasons



- The third visualization illustrates monthly revenue trends from 2021 to 2023, showing clear peaks during certain months. These peaks correspond with major shopping periods, such as holidays and end-of-season sales. For instance, the data shows a spike in revenue around New Year's and back-to-school periods, suggesting strong seasonality in consumer spending."
- To be exact, it captures the fluctuations in monthly revenue, with highlighted peaks during certain periods. This seasonal variation in shopping behavior is important for businesses to understand, as it suggests when to expect high sales and when to prepare for quieter months.
- Note: The sharp decline observed in early 2023 is most likely caused by incomplete reporting for the final month in the dataset, rather than a true decrease in consumer spending. This highlights a limitation of the data collection period and should be interpreted with caution.

Purchase Volume Increases with Age, with Clothing as the Leading Category Across All



- This heatmap shows the number of transactions for each product category across different age groups.
- Clothing is the dominant category across all age groups, confirming its central role in mall spending. Purchase frequency increases with age, with the 56+ group showing the highest engagement across most categories. Younger consumers display lower and more selective purchasing behavior, indicating different marketing opportunities by age segment.
- These patterns indicate that age is a key behavioral driver in shopping malls, influencing product demand beyond overall spending levels. Retail strategies could therefore benefit from age-targeted promotions and category placement within malls.

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